

EE1101: Circuits and Network Analysis - Final Exam

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1. The exam duration is 120 minutes, starting at 10:10. Ensure that you submit both the question paper and the answer book by 12:10. Please be aware that **late submissions** may incur **penalties**.
2. Read all questions carefully. Misinterpretations will **not** be considered. It is essential to write your solutions clearly and legibly in the answer book. In a previous quiz, a few students submitted answers that were not clearly written. Going forward, such responses will **not** be evaluated.
3. Any form of unfair practice will result in an **F grade** and will be reported to the disciplinary committee.

1. (10 points) Consider the circuit shown in Fig. 1(a).

- (a) (4 points) Derive an expression for resonant frequency of the circuit. Compute the resonant frequency for the given values of circuit elements.
- (b) (4 points) For the given values of the circuit elements, compute the quality factor Q of the circuit at resonance.
- (c) (2 points) Compute the bandwidth of the circuit using the bandwidth - quality factor - resonant frequency relationship.

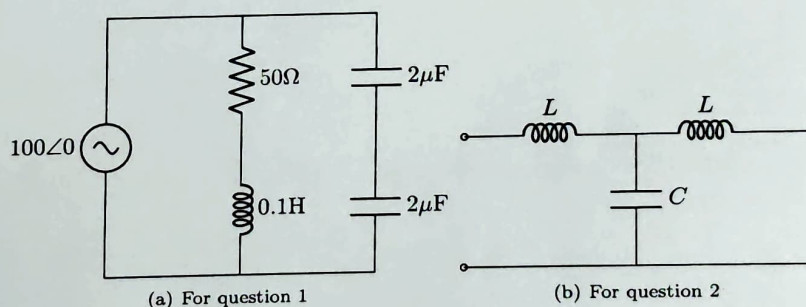


Fig. 1: Circuits for Questions 1 and 2

2. (6 points) Compute the inverse hybrid parameters (g -parameters) of the two-port network shown in Fig. 1(b).
3. (6 points) Consider two two-port networks connected as shown in Fig. 2. You are free to choose any of the standard two-port parameters (Z , Y , h , g , or $ABCD$) to describe the individual two-port networks. For the chosen parameters, derive the overall parameters of the combined two-port network. Clearly indicate the polarity of voltages and currents considered in your analysis.

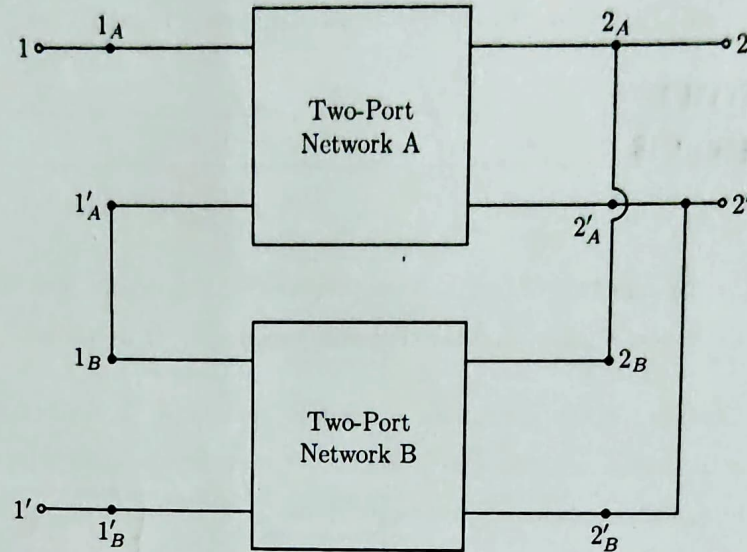


Fig. 2: For question 3

4. (6 points) Consider a two-port network characterized by the admittance parameters. A voltage source in series with a resistor R_1 is connected at the input port. Derive the Thevenin equivalent circuit at the output port of the two-port network.
5. (7 points) In the substitution theorem, we replace an element connected to a linear network with a voltage source or current source such that the voltage across or current through the element remains unchanged. Now, consider a linear network and our objective is to replace an element with a combination of elements such that both the voltage across and current associated with the port remain unchanged. Is this possible? Justify your answer with an example or a counter-example.